First-Order Compactness via Ultraproducts

An End-to-End Meta-Math Walkthrough

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Abstract

Each finite fragment gets its own model. An ultrafilter regards all fragments containing any chosen sentence as large; the quotient product then makes every sentence in the theory true at once. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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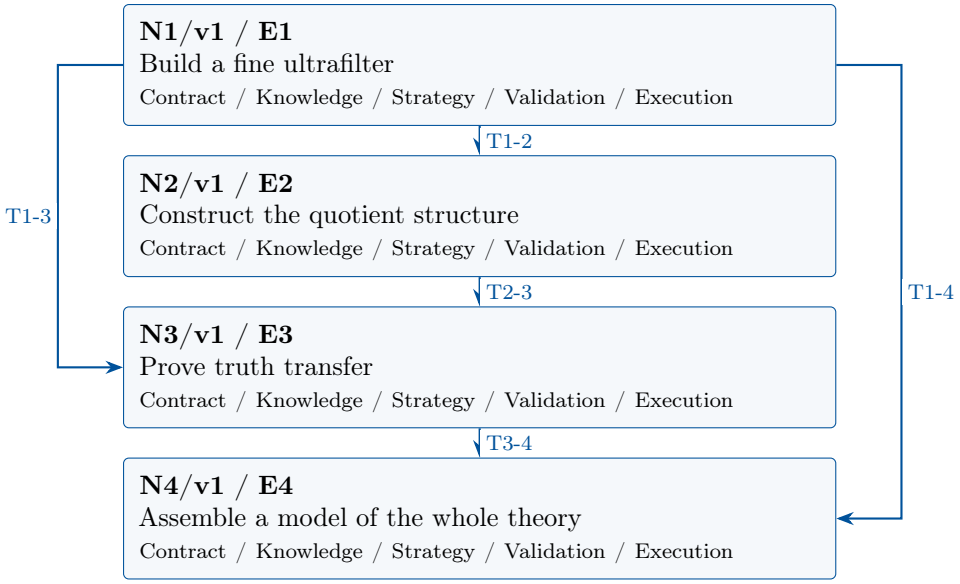
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let T be a set of sentences in a first-order language L with finite-arity relation and function symbols and nonempty structures. If every finite subset of T has a model, prove T has a model. Work in classical ZFC set theory, including choice. The proof constructs an ultraproduct and proves the required truth-transfer lemma rather than importing first-order completeness.*

A. Foundations and permitted imports

Use Zorn's lemma, choice of structures and witnesses from set-indexed nonempty families, finite-intersection filters, and the recursive semantics of first-order formulas. A proper filter on an index set contains the whole set, not the empty set, and is closed under finite intersections and supersets. An ultrafilter is a maximal proper filter. Function symbols may include zero-arity constants. General formulas can be expressed using atomic formulas, negation, conjunction, and existential quantification.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

Receiving activity. N2.1

Compatibility. E1/v1; choose a nonempty model $M_s \models s$ for each $s \in I$. Check: Finite intersections, not countable intersections, suffice for each finite-arity symbol; representatives must not affect truth or function values.

T1-3: N1/E1/v1 to N3. There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1, a first-order formula θ, and product representatives for its finitely many parameters. Check: Existential witnesses may vary with the index. Requiring one common element of all models would be an invalid strengthening.

T2-3: N2/E2/v1 to N3. The product modulo $a \sim b \iff \{s : a_s = b_s\} \in \mathcal{U}$ is a well-defined nonempty L-structure M.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1, a first-order formula θ, and product representatives for its finitely many parameters. Check: Existential witnesses may vary with the index. Requiring one common element of all models would be an invalid strengthening.

T1-4: N1/E1/v1 to N4. There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

Receiving activity. N4.1

Compatibility. E1/v1 cones and E3/v1 truth transfer for sentences. Check: The final assertion quantifies over every sentence separately; no infinite intersection in the ultrafilter is required.

T3-4: N3/E3/v1 to N4. $M \models \theta([a^1], \ldots, [a^r])$ iff $\{s : M_s \models \theta(a_s^1, \ldots, a_s^r)\} \in \mathcal{U}$.

Receiving activity. N4.2

Compatibility. E1/v1 cones and E3/v1 truth transfer for sentences. Check: The final assertion quantifies over every sentence separately; no infinite intersection in the ultrafilter is required.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Build a fine ultrafilter

Object and role. Extend the filter generated by the cones to an ultrafilter and derive its complement decision property. **Local obligation:** There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

Inputs. $I = \{s \subset T : s \text{ finite}\}$ and $C_\varphi = \{s \in I : \varphi \in s\}$ for $\varphi \in T$.

C — Contract

Publish E1/v1 only after establishing this obligation: There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

K — Knowledge

$I = \{s \subset T : s \text{ finite}\}$ and $C_\varphi = \{s \in I : \varphi \in s\}$ for $\varphi \in T$. Retain the stated assumptions and import versions.

S — Strategy

Extend the filter generated by the cones to an ultrafilter and derive its complement decision property.

V — Validation

Every finite intersection of cones contains their finite list of sentences as an index, so it is nonempty.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Finite intersections of the cones are nonempty: the finite union of the required sentences is itself an index. They therefore generate a proper filter (if T is empty use the filter containing I). The union of a chain of proper filters is again a proper filter, because any finitely many of its members occur together in one member of the chain. Zorn supplies a maximal proper filter $\mathcal{U}$.

N1.2 If $A \notin \mathcal{U}$, adjoining it cannot generate a proper larger filter. Thus some $F \in \mathcal{U}$ has $F \cap A = \varnothing$; otherwise its intersections with old filter members would give a proper extension. Therefore $I \setminus A \in \mathcal{U}$. Both complementary sets cannot belong, since their intersection is empty. This is the required decision property.

Export E1/v1. There is an ultrafilter $\mathcal{U}$ on I containing every C_φ, and exactly one of $A, I \setminus A$ belongs to $\mathcal{U}$ for each $A \subset I$.

Acceptance gate. Every finite intersection of cones contains their finite list of sentences as an index, so it is nonempty.

Targeted repair. If maximality is used without a chain upper bound, supply the union-of-chain argument and check that the empty set never enters. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Construct the quotient structure

Object and role. Quotient the product by agreement on an ultrafilter-large set and define symbols coordinatewise. **Local obligation:** The product modulo $a \sim b \iff \{s : a_s = b_s\} \in \mathcal{U}$ is a well-defined nonempty L-structure M.

Inputs. E1/v1; choose a nonempty model $M_s \models s$ for each $s \in I$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The product modulo $a \sim b \iff \{s : a_s = b_s\} \in \mathcal{U}$ is a well-defined nonempty L-structure M.

K — Knowledge

E1/v1; choose a nonempty model $M_s \models s$ for each $s \in I$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Quotient the product by agreement on an ultrafilter-large set and define symbols coordinatewise.

V — Validation

Finite intersections, not countable intersections, suffice for each finite-arity symbol; representatives must not affect truth or function values.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Choice gives the family of models and an element in each domain, so their product is nonempty. Agreement is reflexive and symmetric; transitivity follows because the intersection of two large agreement sets is large and is contained in the third agreement set. Thus $\sim$ is an equivalence relation.

N2.2 Interpret a function on classes by applying it in each M_s, and interpret a relation $R([a^1], \ldots, [a^r])$ as true when $\{s : M_s \models R(a^1_s, \ldots, a^r_s)\} \in \mathcal{U}$. If representatives change, they agree on the finite intersection of their large agreement sets. Function outputs and relation truth agree there, so their quotient value or largeness is unchanged. Equality is the equivalence relation itself; constants are their coordinatewise classes. This defines M.

Export E2/v1. The product modulo $a \sim b \iff \{s : a_s = b_s\} \in \mathcal{U}$ is a well-defined nonempty L-structure M.

Acceptance gate. Finite intersections, not countable intersections, suffice for each finite-arity symbol; representatives must not affect truth or function values.

Targeted repair. If infinitely many representative agreements are intersected for one formula, use only its finite tuple of free variables and the finite arities actually present. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Prove truth transfer

Object and role. Induct on terms and formula construction; use chosen witnesses for the existential direction. **Local obligation:** $M \models \theta([a^1], \ldots, [a^r])$ iff $\{s : M_s \models \theta(a_s^1, \ldots, a_s^r)\} \in \mathcal{U}$.

Inputs. E1/v1 and E2/v1, a first-order formula θ, and product representatives for its finitely many parameters. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

C **Contract**

Publish E3/v1 only after establishing this obligation: $M \models \theta([a^1], \ldots, [a^r])$ iff $\{s : M_s \models \theta(a_s^1, \ldots, a_s^r)\} \in \mathcal{U}$.

K **Knowledge**

E1/v1 and E2/v1, a first-order formula θ, and product representatives for its finitely many parameters. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S **Strategy**

Induct on terms and formula construction; use chosen witnesses for the existential direction.

V **Validation**

Existential witnesses may vary with the index. Requiring one common element of all models would be an invalid strengthening.

E **Execution**

Carry out activities N3.1–N3.3 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 Induction on terms shows evaluation is coordinatewise up to $\sim$: variables and constants have this property by definition, and applying a function preserves it. Atomic equality and relations therefore satisfy the asserted transfer by N2. Negation corresponds to complement, handled by the ultrafilter decision property. Conjunction corresponds to intersection; a finite intersection is large exactly when both factors are large, by filter closure and upward closure.

N3.2 For $\exists z\, \psi(z, \bar{a})$, first suppose M has a witnessing class $[b]$. By the induction hypothesis, $\{s : M_s \models \psi(b_s, \bar{a}_s)\}$ is large. It is contained in the set of indices satisfying the existential formula, so that set is large.

N3.3 Conversely, if the existential satisfaction set is large, choose a witness b_s in each of its models, and choose an arbitrary domain element at the remaining indices. Choice and nonempty domains permit this product element. The set where $\psi(b_s, \bar{a}_s)$ holds contains the large existential set. The induction hypothesis then makes $[b]$ a witness in M. These cases complete the structural induction and prove truth transfer.

Export E3/v1. $M \models \theta([a^1], \ldots, [a^r])$ iff $\{s : M_s \models \theta(a_s^1, \ldots, a_s^r)\} \in \mathcal{U}$.

Acceptance gate. Existential witnesses may vary with the index. Requiring one common element of all models would be an invalid strengthening.

Targeted repair. If the existential case is dismissed as automatic, construct the coordinatewise witnesses and explicitly use the induction hypothesis for the matrix formula. Withhold this export and recheck its descendants after a versioned correction.

D. N4/v1: Assemble a model of the whole theory

Object and role. Use each sentence's cone as a subset of its satisfaction set. **Local obligation:** $M \models T$, proving compactness.

Inputs. E1/v1 cones and E3/v1 truth transfer for sentences. Incoming proof exports: E1/v1 at N4.1, E3/v1 at N4.2.

C — Contract

Publish E4/v1 only after establishing this obligation: $M \models T$, proving compactness.

K — Knowledge

E1/v1 cones and E3/v1 truth transfer for sentences. Incoming proof exports: E1/v1 at N4.1, E3/v1 at N4.2. Retain the stated assumptions and import versions.

S — Strategy

Use each sentence's cone as a subset of its satisfaction set.

V — Validation

The final assertion quantifies over every sentence separately; no infinite intersection in the ultrafilter is required.

E — Execution

Carry out activities N4.1–N4.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N4.1 Fix $\varphi \in T$. Whenever $s \in C_\varphi$, the chosen M_s is a model of s, so it satisfies φ. Therefore $C_\varphi \subset \{s : M_s \models \varphi\}$. The left side is large by N1, hence so is the right side.

N4.2 By N3, $M \models \varphi$. Since this holds for every $\varphi \in T$, $M \models T$. No common index need satisfy all of T; the quotient construction is what supplies the new model.

Export E4/v1. $M \models T$, proving compactness.

Acceptance gate. The final assertion quantifies over every sentence separately; no infinite intersection in the ultrafilter is required.

Targeted repair. If all cones are intersected and assumed large, replace that step by one-sentence-at-a-time truth transfer; filters are only finitely intersection-closed. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Build a fine ultrafilter

1. **N1.1.** Finite intersections of the cones are nonempty: the finite union of the required sentences is itself an index. They therefore generate a proper filter (if T is empty use the filter containing I). The union of a chain of proper filters is again a proper filter, because any finitely many of its members occur together in one member of the chain. Zorn supplies a maximal proper filter $\mathcal{U}$.

2. **N1.2.** If $A \notin \mathcal{U}$, adjoining it cannot generate a proper larger filter. Thus some $F \in \mathcal{U}$ has $F \cap A = \varnothing$; otherwise its intersections with old filter members would give a proper extension. Therefore $I \setminus A \in \mathcal{U}$. Both complementary sets cannot belong, since their intersection is empty. This is the required decision property.

B. N2 — Construct the quotient structure

1. **N2.1.** Choice gives the family of models and an element in each domain, so their product is nonempty. Agreement is reflexive and symmetric; transitivity follows because the intersection of two large agreement sets is large and is contained in the third agreement set. Thus $\sim$ is an equivalence relation.

2. **N2.2.** Interpret a function on classes by applying it in each M_s, and interpret a relation $R([a^1], \ldots, [a^r])$ as true when $\{s : M_s \models R(a_s^1, \ldots, a_s^r)\} \in \mathcal{U}$. If representatives change, they agree on the finite intersection of their large agreement sets. Function outputs and relation truth agree there, so their quotient value or largeness is unchanged. Equality is the equivalence relation itself; constants are their coordinatewise classes. This defines M.

C. N3 — Prove truth transfer

1. **N3.1.** Induction on terms shows evaluation is coordinatewise up to $\sim$: variables and constants have this property by definition, and applying a function preserves it. Atomic equality and relations therefore satisfy the asserted transfer by N2. Negation corresponds to complement, handled by the ultrafilter decision property. Conjunction corresponds to intersection; a finite intersection is large exactly when both factors are large, by filter closure and upward closure.

2. **N3.2.** For $\exists z\, \psi(z, \bar{a})$, first suppose M has a witnessing class $[b]$. By the induction hypothesis, $\{s : M_s \models \psi(b_s, \bar{a}_s)\}$ is large. It is contained in the set of indices satisfying the existential formula, so that set is large.

3. **N3.3.** Conversely, if the existential satisfaction set is large, choose a witness b_s in each of its models, and choose an arbitrary domain element at the remaining indices. Choice

and nonempty domains permit this product element. The set where $\psi(b_s, \bar{a}_s)$ holds contains the large existential set. The induction hypothesis then makes $[b]$ a witness in M. These cases complete the structural induction and prove truth transfer.

D. N4 — Assemble a model of the whole theory

1. **N4.1.** Fix $\varphi \in T$. Whenever $s \in C_\varphi$, the chosen M_s is a model of s, so it satisfies φ. Therefore $C_\varphi \subset \{s : M_s \models \varphi\}$. The left side is large by N1, hence so is the right side.
2. **N4.2.** By N3, $M \models \varphi$. Since this holds for every $\varphi \in T$, $M \models T$. No common index need satisfy all of T; the quotient construction is what supplies the new model.

□

V. WORKED ORCHESTRATION

In the language of ordered fields, add a constant c and all sentences $c > \underbrace{1 + \cdots + 1}_{n \text{ terms}}$ for $n \geq 1$, together with the ordered-field axioms. Every finite fragment has a model in $\mathbb{R}$ by choosing c larger than the finitely many listed integers. Compactness gives an ordered field with an element larger than every standard positive integer. This is a non-Archimedean model, not an infinitely large real number.

A. Boundary and hypothesis audit

Compactness fails if one insists that the resulting model be finite. The sentences "there are at least n distinct elements" have every finite fragment satisfiable in a finite set, but no single finite model satisfies them all. The theorem permits arbitrary nonempty models.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N4 argument: intersect all cones and choose one finite-fragment model satisfying everything. For the increasing-cardinality example that intersection contains no suitable finite index. **Diagnosis:** Strategy used infinite intersection closure that an ultrafilter need not possess. **Repair:** form the quotient product and use the proven truth-transfer lemma separately for each sentence. The original first-order compactness theorem is unchanged; recheck every sentence via its own cone.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use compactness to build models with prescribed finite consistency patterns, keeping nonstandard objects distinct from objects in the original structure.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Andrew Marks, Set Theory notes for sets and ultraproducts; Berkeley, Mathematical Logic for Mathematicians, Part I for first-order semantics; Andrew Marks, Computability Theory notes for effective procedures and undecidability. The proofs below specify their own set-theoretic and semantic assumptions.